

ANALYSIS AND INVESTIGATION OF THE KIBANA'S DASHBOARD

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1.0 INTRODUCTION

A Security Operations Centre (SOC) and a Security Incident and Event Management (SIEM) system are two interconnected approaches that organisations employ to strengthen their cybersecurity efforts. The SOC is dedicated to monitoring and responding to threats, whereas SIEM offers a comprehensive platform for the collection, analysis, and management of security data. A SIEM system integrates Security Information Management (SIM) and Security Event Management (SEM), providing sophisticated tools to oversee log data from multiple sources, identify anomalies, and respond promptly to reduce potential threats.

SIEM technology is essential in contemporary security practices, as it identifies vulnerabilities and addresses them proactively to maintain seamless business operations. These systems allow security teams to identify atypical user behaviours, automate threat detection procedures with AI, and enhance incident response initiatives.

An ELK SIEM solution utilises Elasticsearch, Logstash, and Kibana to provide robust tools for the analysis and visualisation of security data. Organisations can utilise the Kibana dashboard to monitor security events, detect potential risks, investigate threats, and manage escalation processes. The analysis generally involves assessing suspicious IP addresses, identifying threat levels, and measuring the severity of detected incidents to enhance response efficiency and protect the system.

2.0 DASHBOARD OF SIEM

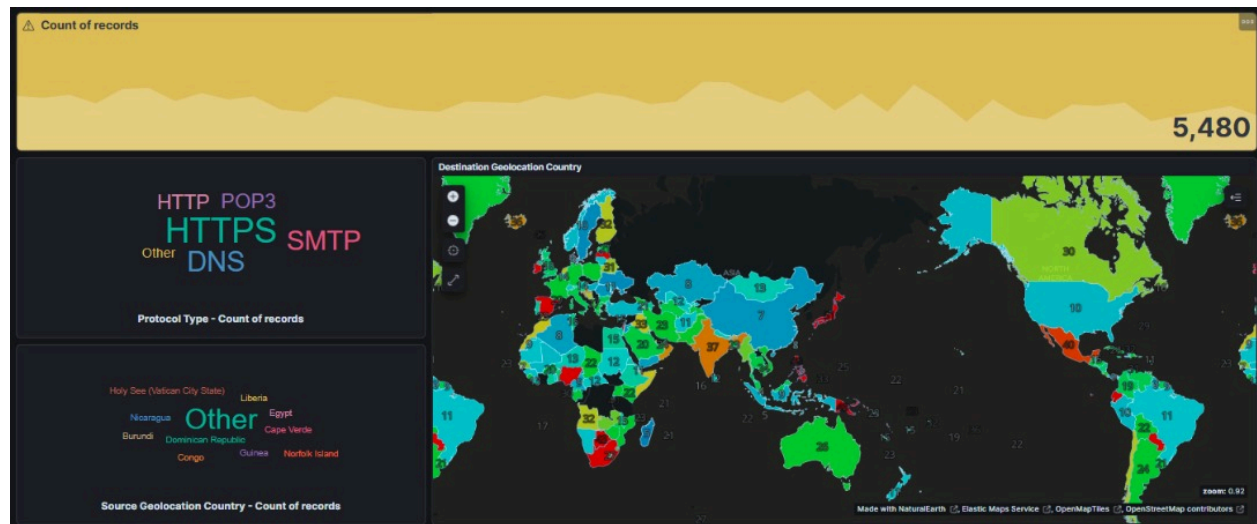


Figure 1: Main Dashboard

This figure represents the main overview of the SIEM system's activities. It includes a summary of all logged events, showing real-time alerts, critical security trends, and key metrics such as the number of threats detected, their severity levels, and the systems affected. This overview serves as the starting point for deeper investigation into specific events or trends.

There are 4 different interfaces in the main dashboard that consist of a count of records, protocol type, source geolocation country, and destination geolocation country. Each of these played a role in information-gathering activity. Each of them is functional and creates an illusion for the viewer to help determine certain information.

- Count of records
- Protocol type
- Source geolocation country
- Destination geolocation country

Count of Records

At first glance, the count of records allows the viewer to see the count of records that are referencing the count of threats detected. In addition, graph features are added to show the escalation and deescalation of event counts while displaying the count of records.

Protocol Type

A small part of dashboards are contributed to the protocol type. This is important to detect patterns in network activity.

Source Destination Country

Source Destination Country shows countries that become a source from a packet transmission where knowing the sender and receiver is crucial for network traffic analysis. The importance of identifying the critical information logs location are to identify the source origins of where the packet are originated from which is the sender IP address. Knowing the sender helps determines the origins of traffic and trace back to the possible issues threat such as DDoS, spam, phishing and suspicious activity.

Destination Geolocation Location

Destination Geolocation Location allows security monitoring which the logging source and destination countries provides insights into some geographic traffic patterns. For example, knowing source and unusual traffic patterns from unexpected and suspicious regions can hints for potential attacks and anomalies, hence locating the destination for the suspicious packets can improve optimization of security implementation such as routing strategies for better security and performance. The sensitive endpoint that are targeted can be enhanced with new policies for protection.

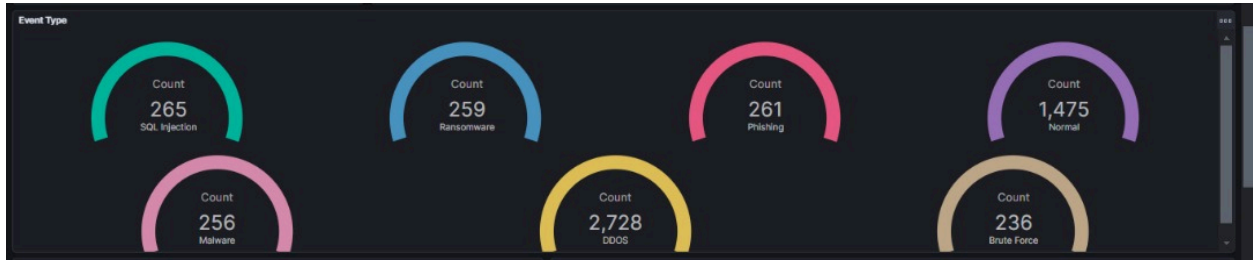


Figure 2: Type Of Malware Detected By Kibana

This figure displays a categorised breakdown of malware detected within the network. Each type of malware is represented, allowing administrators to understand which threats are most prevalent. For example, categories like ransomware, spyware, and Trojans are visualized, making it easy to prioritize responses based on the type and frequency of threats.

Event Type

Summary of Event Type records counts:

Event Type	Counts of Records
SQL Injection	265
Malware	256
Ransomware	259
DDoS	2,728
Phishing	261
Brute-Force	236
Normal	1475

IP Addresses Records Table

IP Address Records Table				
Top 5 values of Source IP Address	Top 5 values of Destination IP Address	Top 5 values of Source Organization	Top 5 values of Destination Organization	Count of records
1.99.99.41	104.244.42.129	Guzman Inc	Hernandez Ltd	11
1.144.155.98	138.68.9.83	Hahn-Chan	Hensley Group	11
2.146.195.152	157.185.179.12	Moss, Schaefer and Simmons	Jordan, Walton and Walker	11
3.63.165.228	185.220.102.241	Castillo Ltd	Clayton, Richards and Hamilton	11
4.154.46.41	216.239.38.21	Thomas LLC	Thompson Inc	11
Other	183.136.225.32	Jones Group	Carlson-Raymond	11
Other	183.136.225.32	Jones Group	Avila-Bailey	11
Other	183.136.225.32	Anderson, Davis and Howard	Moore-Sanchez	11
Other	183.136.225.32	Bowman and Sons	Strong Group	11
Other	183.136.225.32	Cruz LLC	George-Reed	11
Other	183.136.225.32	Dixon, Chambers and Baird	Goodman-Underwood	11

Figure 3: IP Addresses record for each type.

This figure provides a list of IP addresses linked to specific threat categories. It allows administrators to trace malicious activities back to their sources. For example, if a specific IP address is associated with ransomware, the figure highlights the connection. This is crucial for identifying patterns and taking targeted action, such as blocking the IP.

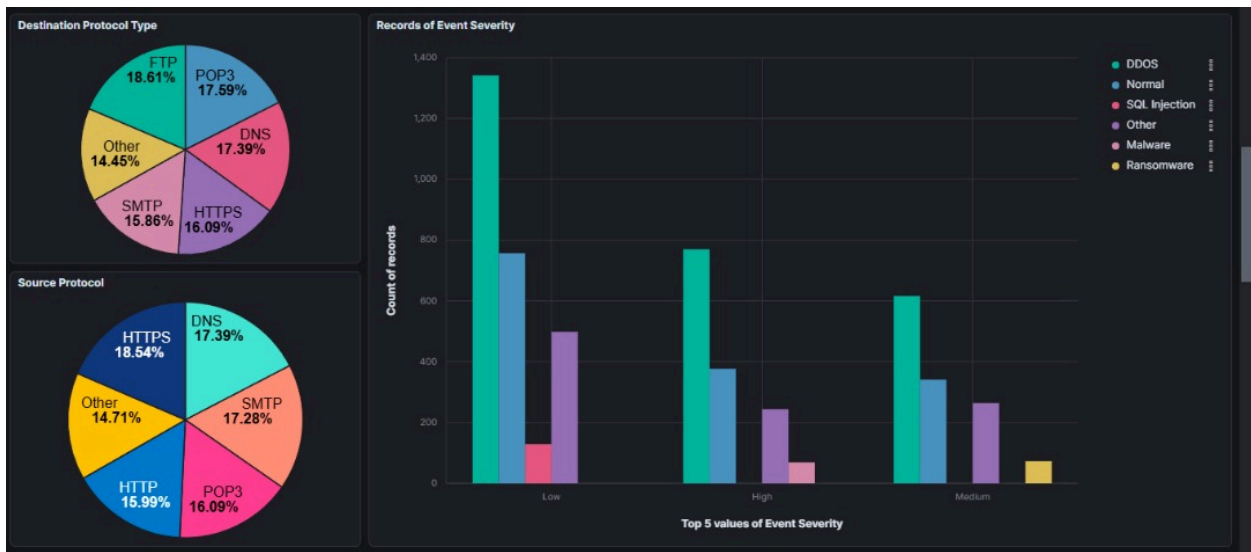


Figure 4: A Pie Chart Of Source And Destination Protocol used and Event Severity of Each Protocol

This figure illustrates the protocols used for source and destination communications within the network, represented as a pie chart. It also categorizes the severity of events associated with each protocol, such as HTTP, HTTPS, or FTP. By visualizing this data, administrators can identify protocols commonly exploited by attackers and take preventive measures to secure them.

Destination & Source Protocol Type

- Destination Protocol Type
 - **FTP** - 18.61%
 - **POP3** - 17.59%
 - **DNS** - 17.39%
 - **HTTPS** - 16.09%
 - **SMTP** - 15.86%
 - **OTHER** - 14.45%
- Source Protocol Type
 - **HTTPS** - 18.59%
 - **DNS** - 17.39%
 - **SMTP** - 17.28%
 - **POP3** - 16.09%
 - **HTTP** - 15.99%
 - **Other** - 14.71%

Record of Event Severity

Based on the multi-bar graphs, shows that event severity recorded are divided into multiple section from high to low severity. Colors are used to determine the event type while from the leftmost section are the bars that refers to high severity events to rightmost section refers to low severity.

Each of the colors represents:

Green - DDOS

Blue - Normal

Red - SQL Injection

Purple - Other

Orange - Malware

Yellow - Ransomware

Median of Source & Destination Port with Longitude and Latitude Source and Destination

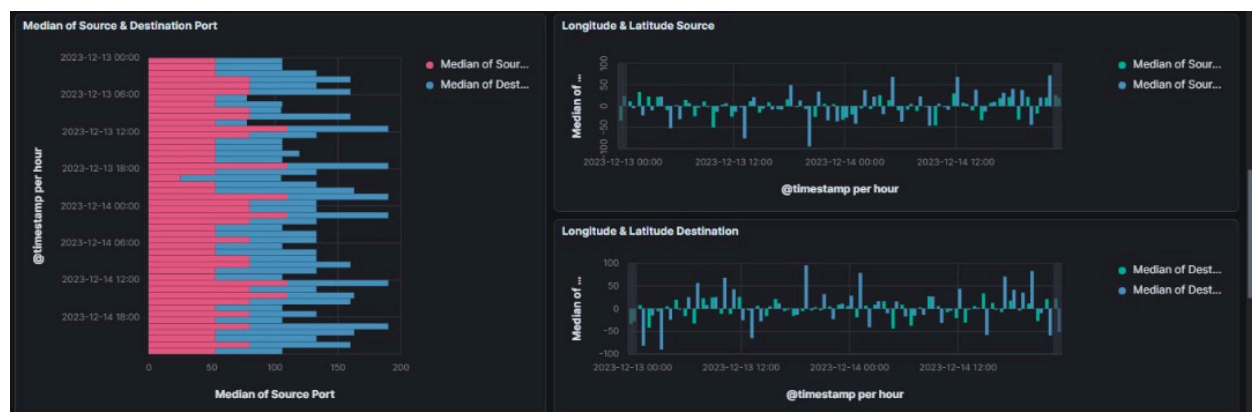


Figure 5: Bar And Graphical Chart for the Geolocation with Longitude and Latitude of Destination and Source

This figure shows the geographical distribution of source and destination IP addresses based on their longitude and latitude. It provides a visual representation of where network traffic is

originating and terminating. By analyzing this data, security teams can identify potential hotspots for malicious activity or determine if unusual geographic patterns indicate an ongoing attack.

Destination & Source Geolocation City



Figure 6: Average City with Affiliation with Destination and Source IP Address.

This figure highlights the cities associated with the source and destination IP addresses. It aggregates data to show the most frequent city connections, which helps identify if certain regions are more prone to generating malicious traffic. For example, if a majority of threats originate from a specific city, it may indicate the presence of a threat actor or compromised systems in that region.

3.0 INVESTIGATION AND RESEARCH FINDINGS

To investigate an event, we can use websites such as VirusTotal. The relevancy of investigating is to primarily gather information about the source, location, and potential risk associated with a specific IP address. Whereas each platform uses cases that are different but are beneficial to the network investigation process, such as VirusTotal to analyse suspicious files, URLs, and IP addresses.

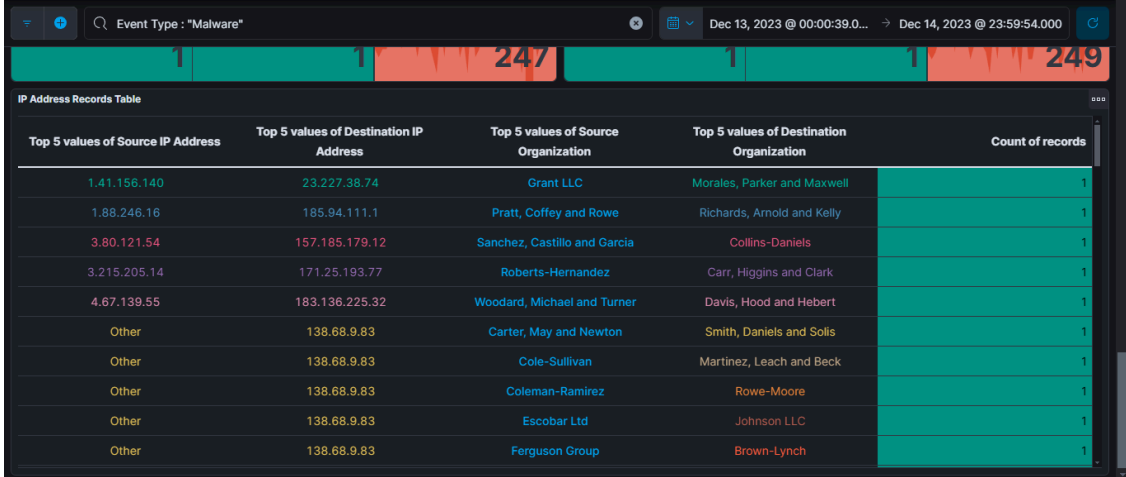


Figure: VirusTotal Websites for analysing the IP activities

VirusTotal is an online platform that specialises in analysing and identifying suspicious files, URLs, IP addresses, and domains for malware and malicious software activities. There are a few main key functions that allow this platform to be constantly used and trusted by most organisations to be used for suspicious activity detection. Those key functions include malware analysis, threat intelligence, and, most importantly, investigation for suspicious and malicious activities.

3.1. TRUE-POSITIVE

A true-positive result in a Security Incident and Event Management (SIEM) system is when a tool correctly identifies a threat or vulnerability. For example, if a suspicious activity from a specific IP address is flagged as malicious by the SIEM and further investigation confirms the activity as a genuine threat, this is considered a true positive. This accurate detection allows the security team to take timely actions such as blocking the IP or implementing additional security measures. True-positive alerts strengthen the system's reliability and ensure effective threat management.



The screenshot displays a SIEM interface with a search bar at the top showing "Event Type: 'Malware'". Below the search bar, there are two large red bars with the numbers "247" and "249". The main content area is titled "IP Address Records Table" and contains a table with the following data:

Top 5 values of Source IP Address	Top 5 values of Destination IP Address	Top 5 values of Source Organization	Top 5 values of Destination Organization	Count of records
1.41.156.140	23.227.38.74	Grant LLC	Morales, Parker and Maxwell	1
1.88.246.16	185.94.111.1	Pratt, Coffey and Rowe	Richards, Arnold and Kelly	1
3.80.121.54	157.185.179.12	Sanchez, Castillo and Garcia	Collins-Daniels	1
3.215.205.14	171.25.193.77	Roberts-Hernandez	Carr, Higgins and Clark	1
4.67.139.55	183.136.225.32	Woodard, Michael and Turner	Davis, Hood and Hebert	1
Other	138.68.9.83	Carter, May and Newton	Smith, Daniels and Solis	1
Other	138.68.9.83	Cole-Sullivan	Martinez, Leach and Beck	1
Other	138.68.9.83	Coleman-Ramirez	Rowe-Moore	1
Other	138.68.9.83	Escobar Ltd	Johnson LLC	1
Other	138.68.9.83	Ferguson Group	Brown-Lynch	1

Figure: Report addresses functions for malicious and suspicious IP activities

Shown in the figure is a list of source IP addresses that are derived from the Security Information and Event Management (SIEM) system. To prove there is a source IP address resulting in a true-positive situation, a source IP address is selected for checking and inspection using trusted tools among the network administrators, such as VirusTotal. The specific IP address chosen for inspection and investigation is 185.94.111.1.

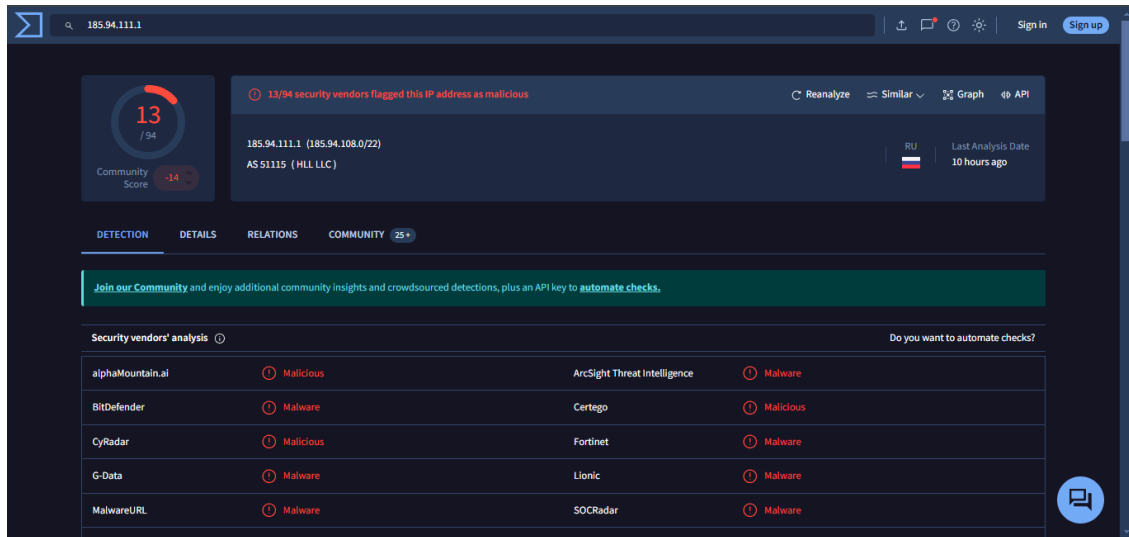
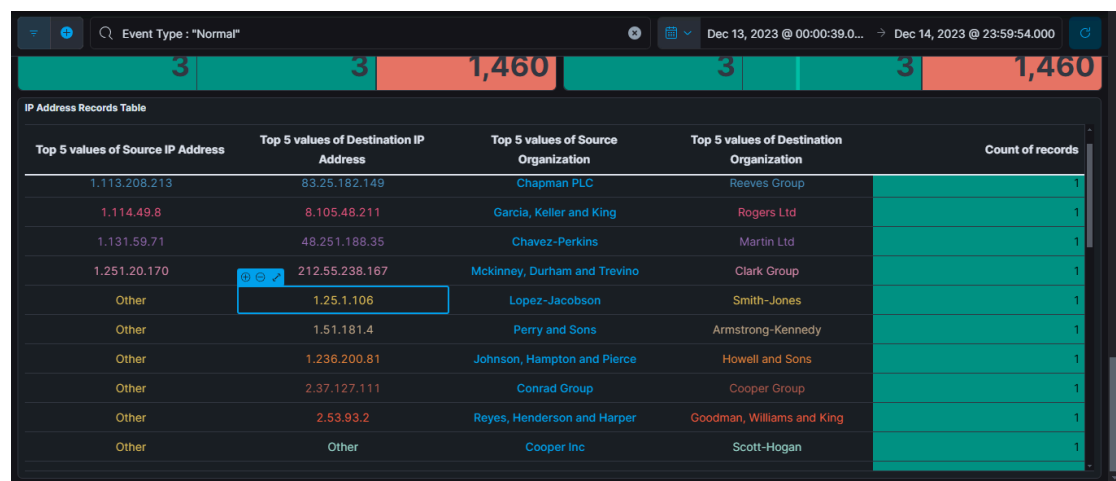


Figure:

The VirusTotal online platform is used to detect malware and malicious software activities within the source IP address 185.94.111.1. Referring to the figure shows that the specific IP address confirmed to be involved in malware and malicious activities. This highlights a consistent pattern of malicious behaviour linked to this IP address. The behaviour aligns with the category of threats flagged by the SIEM system, as displayed in the Kibana interface, where similar IP addresses have been filtered and categorised as malware. To summarise, the specific IP addresses are proven to be true positives since the malicious activities flagged by VirusTotal align with the SIEM system.

3.2. TRUE-NEGATIVE

In an SIEM system, a true-negative result occurs when the system correctly identifies non-threatening or legitimate activities as safe. For instance, if an IP address is flagged for analysis but further investigation through tools like VirusTotal indicates no signs of malicious activity, the system correctly categorises it as benign. This prevents unnecessary investigations and ensures that legitimate operations proceed without interruptions, contributing to the efficiency of the security team.



The screenshot shows the Kibana interface for the 'IP Address Records Table'. The table displays the top 5 values of Source IP Address, Top 5 values of Destination IP Address, Top 5 values of Source Organization, and Top 5 values of Destination Organization, along with the Count of records. The IP address 1.25.1.106 is highlighted in the 'Top 5 values of Destination IP Address' column.

Top 5 values of Source IP Address	Top 5 values of Destination IP Address	Top 5 values of Source Organization	Top 5 values of Destination Organization	Count of records
1.113.208.213	83.25.182.149	Chapman PLC	Reeves Group	1
1.114.49.8	8.105.48.211	Garcia, Keller and King	Rogers Ltd	1
1.131.59.71	48.251.188.35	Chavez-Perkins	Martin Ltd	1
1.251.20.170	212.55.238.167	McKinney, Durham and Trevino	Clark Group	1
Other	1.25.1.106	Lopez-Jacobson	Smith-Jones	1
Other	1.51.181.4	Perry and Sons	Armstrong-Kennedy	1
Other	1.236.200.81	Johnson, Hampton and Pierce	Howell and Sons	1
Other	2.37.127.111	Conrad Group	Cooper Group	1
Other	2.53.93.2	Reyes, Henderson and Harper	Goodman, Williams and King	1
Other	Other	Cooper Inc	Scott-Hogan	1

Figure 7: List of normal IP Addresses in Kibana

Shown in the figure is a list of normal IP addresses recorded by the SIEM system and displayed in Kibana. The listing consists of IP addresses flagged as normal behaviour and activities that become benchmarks for identifying deviations. These IP addresses represent trusted sources and destinations within the network, helping to distinguish between normal and potential threat vectors in the network traffic. Accordingly, the SIEM system helps reduce the unnecessary inspection and investigation and allows the network security administrators team to focus on real threats. The specific IP address chosen for inspection and investigation is 1.25.1.106.

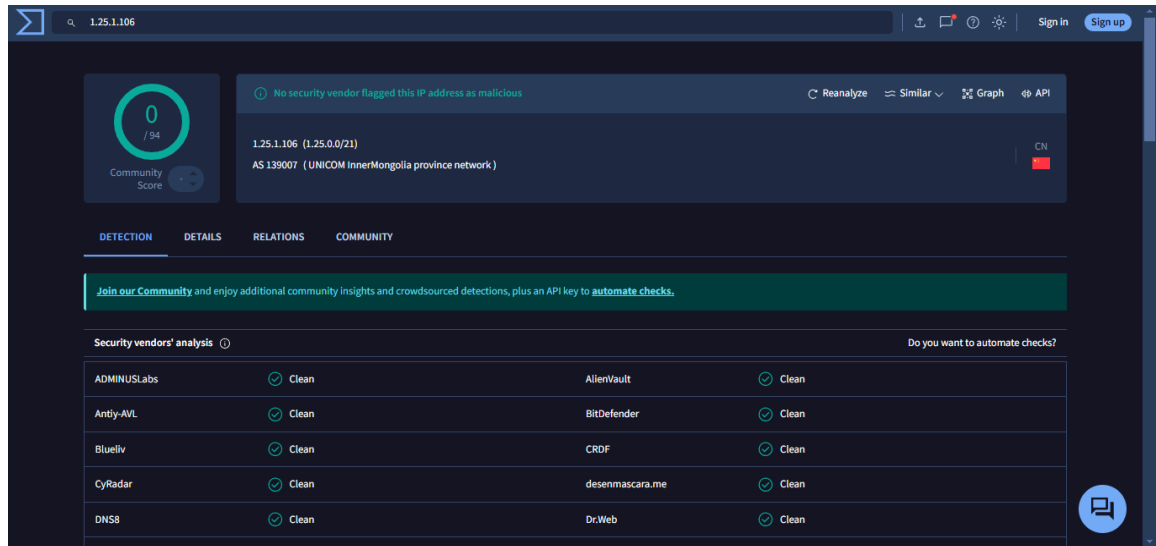
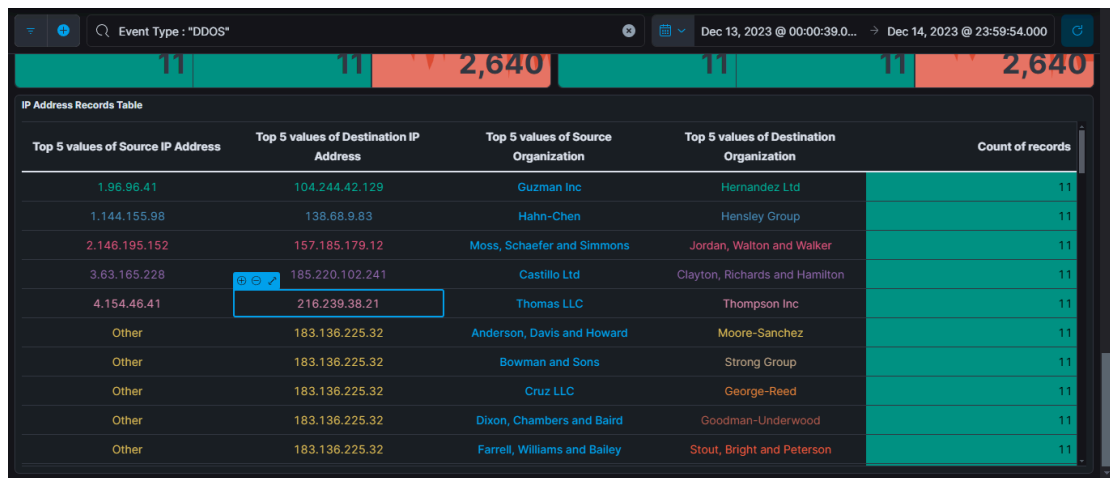


Figure:

Referring to the figure, a specific IP address is in the process of inspection and investigation for its activity. Based on the SIEM system, the IP address should be involved in normal and non-threatening activities in the network traffic. By comparison, the results indicate that the results of VirusTotal inspection prove that the results of the SIEM system align with the results of VirusTotal inspection, where both platforms correctly identify the specific IP address as normal and non-abusive. Hence, it is a safe IP address correctly flagged as normal by the SIEM system, displayed in the Kibana interface. The specific IP address is a true negative.

3.3. FALSE-POSITIVE

A false positive occurs when the SIEM system incorrectly identifies a legitimate activity as malicious. For example, an IP address might be flagged as associated with ransomware, but deeper analysis reveals it as legitimate. Such scenarios lead to wasted resources, as the security team must investigate the non-issue, causing potential delays in addressing real threats. Minimising false positives is crucial for maintaining productivity and ensuring the credibility of the system.



The screenshot displays a SIEM interface with a search bar at the top showing "Event Type: 'DDOS'". Below the search bar, there are two summary bars: a green one with "11" and a red one with "2,640". The main table is titled "IP Address Records Table" and contains the following data:

Top 5 values of Source IP Address	Top 5 values of Destination IP Address	Top 5 values of Source Organization	Top 5 values of Destination Organization	Count of records
1.96.96.41	104.244.42.129	Guzman Inc	Hernandez Ltd	11
1.144.155.98	138.68.9.83	Hahn-Chen	Hensley Group	11
2.146.195.152	157.185.179.12	Moss, Schaefer and Simmons	Jordan, Walton and Walker	11
3.63.165.228	185.220.102.241	Castillo Ltd	Clayton, Richards and Hamilton	11
4.154.46.41	216.239.38.21	Thomas LLC	Thompson Inc	11
Other	183.136.225.32	Anderson, Davis and Howard	Moore-Sanchez	11
Other	183.136.225.32	Bowman and Sons	Strong Group	11
Other	183.136.225.32	Cruz LLC	George-Reed	11
Other	183.136.225.32	Dixon, Chambers and Baird	Goodman-Underwood	11
Other	183.136.225.32	Farrell, Williams and Bailey	Stout, Bright and Peterson	11

Figure :

Shown in the figure is a list of source IP addresses that are derived from the Security Information and Event Management (SIEM) system. To prove there is a source IP address resulting in a false-positive situation, a source IP address is selected for checking and inspection using trusted tools among the network administrators, such as VirusTotal. The specific IP address chosen for inspection and investigation is 216.239.38.21

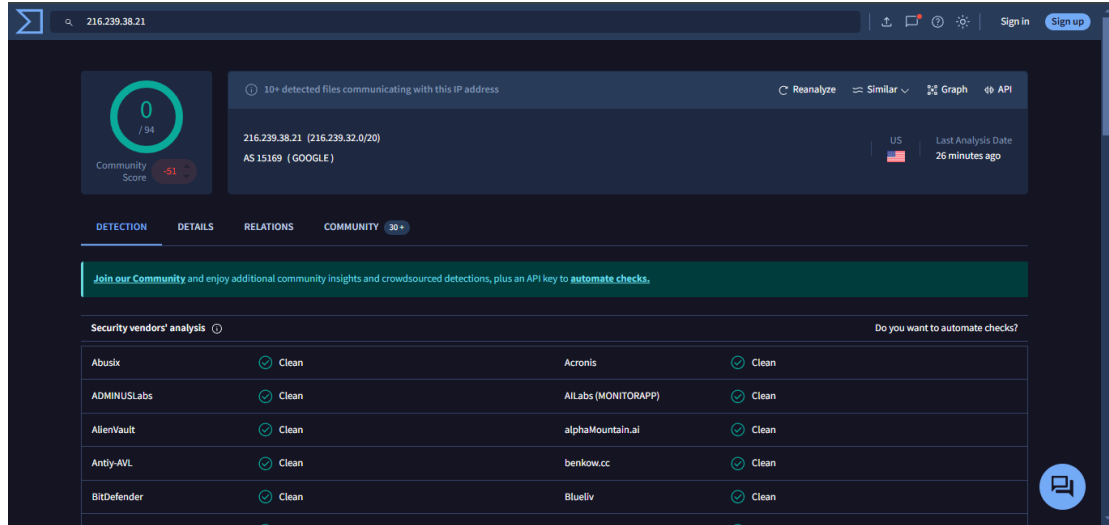


Figure :

The SIEM system flagged the IP address 216.239.38.21 as associated with a Distributed Denial of Service (DDoS) attack. Based on Kibana, this specific IP address should be flagged as DDoS, but further checking proves the IP difference in behaviour. Upon cross-indexing the flagged IP address with VirusTotal and the SIEM system, it was marked clean by all major security vendors in the results or reports. It shows no further details and evidence of abusive activity or any association with the DDoS attacks. Since the flagged IP address does not show any actual or current malicious activity in the VirusTotal analysis, the SIEM system's alert can be categorised as a false positive.

3.4. FALSE-NEGATIVE

A false negative occurs when the SIEM system fails to detect a real threat. For instance, an IP address associated with malicious activity might bypass the SIEM filters and appear as normal. This is a critical failure as it exposes the system to potential risks, making it necessary to refine the SIEM's detection algorithms and ensure comprehensive threat coverage. False negatives are particularly dangerous because they can lead to undetected data breaches or other cyber incidents.

As per inspections that are executed manually by entries of every source IP address, there is none that fulfil the condition of being false-negative. This suggests that SIEM systems that were implemented managed to filter and flag every threat and can efficiently detect real threats.

4.0 CONCLUSIONS

Examining IP addresses plays a crucial role in network security, as it facilitates the identification and mitigation of potential threats. By conducting thorough investigations, we can pinpoint suspicious IP addresses, uncover unauthorised access attempts, and reduce the risk of potential cyberattacks. A prompt investigation enables the establishment of strong security measures and anticipatory threat management.

Utilising tools such as Kibana, VirusTotal, and AbuseIPDB allows for effective analysis and categorisation of events. This improves our comprehension of the various types of alerts and allows for more informed decision-making regarding network activities. Engaging in activities such as customising dashboards and analysing event logs enhances practical skills and strengthens our capacity to recognise suspicious patterns.

In summary, incorporating investigative practices and tools into cybersecurity workflows is essential for reducing risks and preserving the integrity of a network. Practical analysis equips individuals with the knowledge needed to successfully address real-world security challenges, contributing to a more secure digital environment.

5.0 ESCALATION REPORT

Formal Report: Protocol Type Analysis and Escalation

Date: 21/1/2025

Prepared By:

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OBJECTIVE

The report that we are writing here today aims to escalate the findings of protocol types used by threat actors in our investigation of the abnormal behaviour from various IP Addresses. The analysis categorizes threats by their protocol type, and using in-house software, developed by our staff, we are able to analyze which IP Addresses have been flagged by VirusTotal using their Application Program Interface, or API. With this in mind, we can provide insights into their characteristics and potential impact on the respective destinations.

THREATS BASED BY PROTOCOL TYPES USED

The following protocol types have been identified during the analysis are as follow:

1. Distributed Denial Of Service (DDOS)

- **Protocol Used:** DNS, HTTPS, AND HTTP
- **Reasoning for Suspicion:** These Protocols are frequently used to target by utilizing overwhelming service requests in DDOS Attacks.

2. SQL Injection

- **Protocol Used:** HTTP and HTTPS
- **Reasoning for Suspicion:** These protocols are responsible for communication for web-based applications, making them a primary target for SQL injection threats.

3. Malware

- **Protocol Used:** SMTP, POP3, and FTP
- **Reasoning for Suspicion:** Malware is often distributed via Mail Protocols, or by email which are **Simple Mail Transfer Protocol (SMTP)** or **Post Office Protocol (POP3)**, or even through file uploads and downloads, **File Transfer Protocol(FTP)**.

4. Ransomware

- **Protocol Used:** SMTP, HTTP, and HTTPS
- **Reasoning for Suspicion:** Ransomware often is delivered via Mail Protocol, it is almost the same method as malware, but it draws the line of similarity when it also embeds itself in malicious links on web pages.

RECOMMENDATIONS FOR MITIGATION BY PROTOCOL

1. SMTP

- Implement Spam Filters to block Phishing attempts.
- Use Email Encryption to detect malicious payloads from threat actors.

2. HTTPS/HTTP (WEB)

- Install or Implement Web Application Firewalls (WAF)
- Regularly Patch Web Servers and Applications.
- Enable strict transport security.

3. DNS (NETWORK)

- Instil a DNS filtering to block malicious domains or IP addresses.
- Monitor the DNS queries for unusual activity.

4. FTP/POP3 (FILE TRANSFER)

- Enforce Secure File Transfer Protocol
- Scan all incoming files for malware

FURTHER INVESTIGATION FOUND

By Utilizing the VirusTotal API, we can separate the legitimate and illegitimate IPs that request connection to our domain. From the testing itself, the project, handled under our department, utilized the personal API key, which is limited to 500 lookups per day. An enterprise-level of

subscription can avoid this hassle and bottleneck. Further Research and Development are needed to make the software itself real-time and user-friendly towards the staff in our respective department. The software itself is in beta testing, in which the output is quite rough and isn't refined towards the corporate standard. The output is as shown:-

```
Scanning IP (19): 64.241.202.234
Result for 64.241.202.234: Not Malicious
Scanning IP (20): 177.212.76.69
Result for 177.212.76.69: Not Malicious
Scanning IP (21): 157.185.179.12
Result for 157.185.179.12: Malicious (2 reports)
Scanning IP (22): 185.94.111.1
Result for 185.94.111.1: Malicious (13 reports)
```

After further research, it can be integrated towards the Kibana by utilising the VirusTotal API. Further time and Development are needed to make a proof of concept.

CONCLUSION

The findings highlight the incapability of handling such threats from either the local geographical location or outside of its proximity. Due to this, it is in need for proactive measures to address protocol-based threats. Immediate action, acquisitions of resources and agreement with the top departments to implement the recommendations can significantly reduce the risk and impact of future incidents.

Approval of Escalation: Mr. Kanye West

6.0 REFERENCES

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